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WORK EXPERIENCE

Volunteer / Full Time Research Assistant, Singapore University of Technology & Design (SUTD) - ROAR Lab Aug 2022 – May 2024

- Developed perception software and processed 3D LiDAR data for autonomous mobile robots, designing architectures integrating Embedded C++, ROS, and sensor fusion.
- Built a specialized predictive safety model to detect a stair-climbing robot's fall risk, deploying Long Short-Term Memory networks (LSTMs) to improve physical safety metrics by 50%.
- Led the design and development of an intelligent lighting control system for an autonomous vehicle, integrating it seamlessly with the main navigation system via RTOS and APIs.

Student Research Engineer, Singapore Institute of Technology (SIT) Aug 2024 – May 2025

- Designed and implemented an experimental temperature chamber to simulate human body heat signatures and extreme physical environments for environmental robotics testing.
- Engineered a closed-loop, thermistor-based control system to regulate precise environmental variables, ensuring high-fidelity data collection.
- Established repeatable testing and data collection protocols for faculty-led embodied AI and robotics hardware projects.

Machine Learning & Fullstack Engineer (Intern), KLASS Engineering and Solutions Sep 2025 – Aug 2026

- Reproduced and benchmarked state-of-the-art vision and audio AI models (NSG-VD, D3, ReStraV, Speech-Forensics) to evaluate real-time transparency and explainability in threat detection against proprietary pipelines.
- Integrated advanced optical flow (RAFT), segmentation (SAM 2), and point-tracking algorithms (CoTracker) to analyze physical motion derivatives in video sequences for mission-critical data.
- Engineered a comprehensive data generation pipeline linking video extension models (Wan2.2), voice cloning (Chatterbox), and lip-synchronization (LatentSync) to curate a balanced 1,200-video dataset for benchmarking.
- Managed GPU-accelerated experimentation on shared SLURM clusters (NVIDIA H200 nodes) via Docker containers, and deployed fullstack architectures (React, Zustand, FastAPI, WebSockets) for multi-model AI endpoints.
- Designed scalable dataset curation workflows with custom validation splits and lightweight sanity-check protocols before full-scale benchmarking.

PROJECTS

1. Autonomous UAV Navigation in GPS-Denied Environments

- Formulated engineering architectures for indoor UAV navigation utilizing Visual-Inertial Odometry (VIO), spatial AI depth cameras (Luxonis OAK-D Pro), and 1D Time-of-Flight sensors mapped across Raspberry Pi and Pixhawk/ArduPilot controllers.

2. Aeroacoustic Stealth for Unmanned Systems

- Investigated noise reduction strategies for UAVs—combining toroidal propellers, biomimetic owl-wing serrated blade designs, low-KV motors, and Field Oriented Control ESCs—to suppress acoustic signatures.

3. Autonomous Trolley Collection Robot

- Developed an autonomous robot for collecting stray trolleys using an STM32F7 microcontroller with RTOS, implementing tasks for obstacle detection, joystick control, and a custom serial communication protocol.

4. Custom 3D Printer Development

- Built a custom 3D printer, incorporating off-the-shelf modules and developing custom firmware in C++ (Marlin), increasing reliability and accuracy by 50%.

5. Autonomous Vehicle Intelligent Lighting System Aug 2022 – May 2024

- Led the design and development of an intelligent lighting control system for an autonomous vehicle, integrating it seamlessly with the main navigation system via RTOS and APIs.

6. Experimental Thermal Chamber for Embodied AI Testing Aug 2024 – May 2025

- Designed and implemented an experimental temperature chamber to simulate human body heat signatures and extreme physical environments for environmental robotics testing.
- Engineered a closed-loop, thermistor-based control system to regulate precise environmental variables, ensuring high-fidelity data collection.

SKILLS / COMPETENCIES

Robotics & Autonomous Systems

1. Embodied AI
2. ROS
3. SLAM (3D LiDAR)
4. Visual-Inertial Odometry (VIO)
5. Sensor Fusion
6. RTOS
7. PID Control
8. UAV Aeroacoustics
9. PX4 Autopilot
10. ArduPilot
11. Pixhawk

EDUCATION

DigiPen Institute of Technology & Singapore Institute of Technology Aug 2023 – May 2027

- B.Eng. Mechatronics Systems
- Relevant Coursework: Mechanics, Control Systems, Robotics, Embedded Systems, Machine Learning, Systems and Software Engineering

Ngee Ann Polytechnic Sep 2022 – Sep 2023

- Specialist Diploma in Robotics Engineering (SGInnovate PowerX)
- Relevant Coursework: Autonomous Robot Navigation, Machine Learning/Deep Learning for Robotics, Computer Vision

Singapore Polytechnic Apr 2017 – Sep 2020

- Diploma in Mechatronics and Robotics
- Internship grade of B+; coursework in Microcontroller Applications, PLCs, and Robotics Integration

AWARDS / ACHIEVEMENTS / CERTIFICATES

Award / Certificate	Date
MOE (Ministry of Education) Instructor Certification (Achieved at age 18)	Valid Oct 2018 – Oct 2026
Google Crash Course on Python	
Google The Bits and Bytes of Computer Networking	
Coursera Machine Learning	